

MPPT-CONTROLLED GRID-CONNECTED SOLAR PV SYSTEM

DISTRIBUTION AUTOMATION AND SMART GRID (EC9130)



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1. AIMS OF THE 13TH WEEK

This week focused on three core technical milestones to advance the MPPT-Controlled Grid-Connected Solar PV System towards full functionality:

1. Inverter Control – Complete PLL integration for precise grid synchronization.
2. DC-Link Voltage Control – Implement outer voltage PI loop to regulate DC bus voltage at 600 V.
3. Current Loop Control (CLC) – Implement and test CLC to synchronize inverter current with grid.
4. MPPT Optimization – Enhance the P&O algorithm for faster response under dynamic conditions.

2. INVERTER CONTROL – PLL INTEGRATION

The Phase-Locked Loop (PLL) plays a crucial role in synchronizing the inverter output with the grid voltage and frequency. Its primary objective is to ensure that the inverter generates AC power that is in-phase with the grid, ensuring efficient power injection and grid stability.

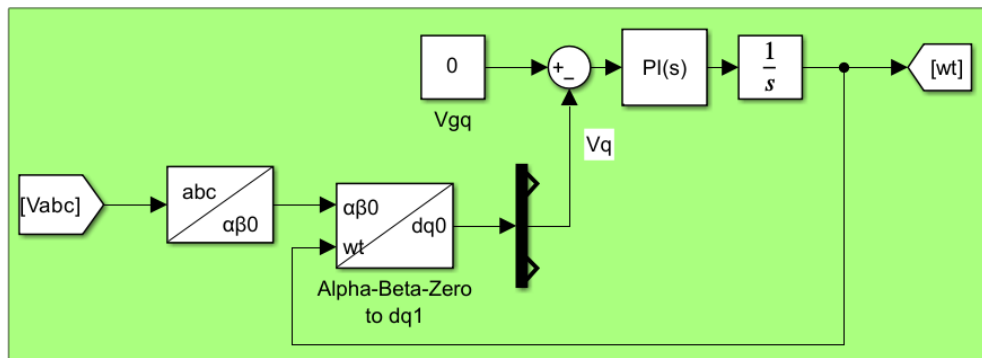


Figure 01: Phase-Locked Loop (PLL) Integration for Grid Synchronization

The PLL integrates the following steps:

- Clarke Transformation (abc to $\alpha\beta 0$): The first step involves transforming the three-phase grid voltage (V_{abc}) into a two-phase stationary reference frame ($\alpha\beta 0$) using the Clarke transformation.
- dq0 Transformation: After converting to the $\alpha\beta 0$ frame, the voltage is further transformed into the rotating dq0 reference frame. This allows the system to align the inverter's frequency and phase with the grid.
- PLL Operation: The PLL adjusts the frequency and phase of the inverter output based on the phase difference between the inverter's output and the grid, ensuring synchronization at all times.

The Phase-Locked Loop (PLL) is essential for ensuring synchronization between the inverter's AC output and the grid, which is crucial for grid stability, especially during voltage sags and load fluctuations. By aligning the inverter's voltage and frequency with the grid, the PLL ensures optimal power injection and reduces the risk of instability.

However, real-time synchronization faces challenges, particularly with voltage sags or rapid load changes, which can disrupt phase alignment. Harmonics and grid faults also complicate accurate phase tracking, requiring precise tuning of the PLL and additional fault detection mechanisms for stable operation.

In summary, while PLL is effective for grid synchronization, overcoming the challenges of dynamic grid conditions and harmonics is key to maintaining power quality and system stability.

3. DC LINK CONTROL

The DC Link Control system ensures that the DC link voltage (V_{dc}) remains stable and at the desired reference value (V_{dc_ref}) for optimal inverter operation. It ensures proper voltage levels are maintained for efficient DC to AC conversion by the inverter.

The DC Link Control system works by comparing the actual DC link voltage (V_{dc}) to the desired reference voltage (V_{dc_ref}). The control algorithm adjusts the system to maintain the desired voltage using a Proportional-Integral (PI) controller.

Table 01: *DC Link Voltage Control Parameters*

Parameter	Description
Reference Voltage (V_{dc_ref})	Set at a constant value of 600 V (as implemented in simulation). This represents the desired DC voltage for efficient inverter operation.
Error Calculation	The error between the reference voltage ($V_{dc_ref} = 600$ V) and the actual DC link voltage (V_{dc}) is calculated at each time step.
PI Controller	The error signal is processed by the PI controller, which adjusts the system output to minimize the error and maintain V_{dc} close to V_{dc_ref} .

The figures below show the MATLAB/Simulink implementation of the DC Link Control subsystem.



Figure 02: *V_{dc_ref} constant block set to 600 V*

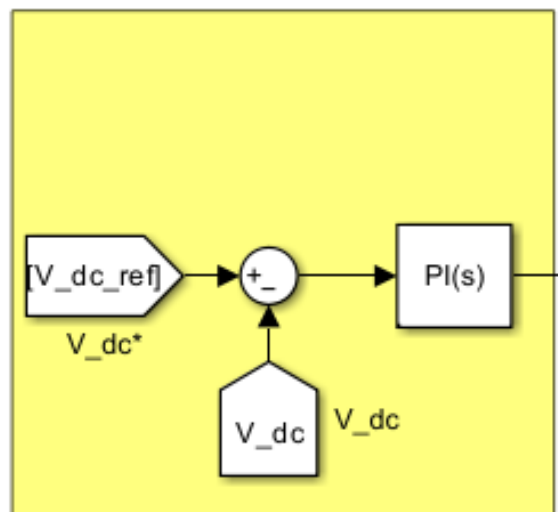


Figure 03: *DC Link Control subsystem*

Significance

- Ensures that the DC voltage is regulated and stable, providing the correct input voltage for the inverter.
- Maintains optimal power conversion from DC to AC with minimal losses.
- The PI controller continuously corrects deviations, ensuring the DC link voltage remains at 600 V under varying solar irradiance and load conditions.
- Stable DC-link voltage directly improves the quality of the AC output and reduces Total Harmonic Distortion (THD).

4. CURRENT LOOP CONTROL (CLC) IMPLEMENTATION

The Current Loop Control (CLC) system is integral to ensuring that the inverter current aligns precisely with the grid current, which is essential for unity power factor operation. The CLC manages both the active power (d-axis current) and reactive power (q-axis current) components, regulating them through a d-q transformation. By adjusting the inverter's output current in real-time, CLC ensures optimal power quality, minimizes reactive power, and guarantees stable and efficient energy transfer from the inverter to the grid.

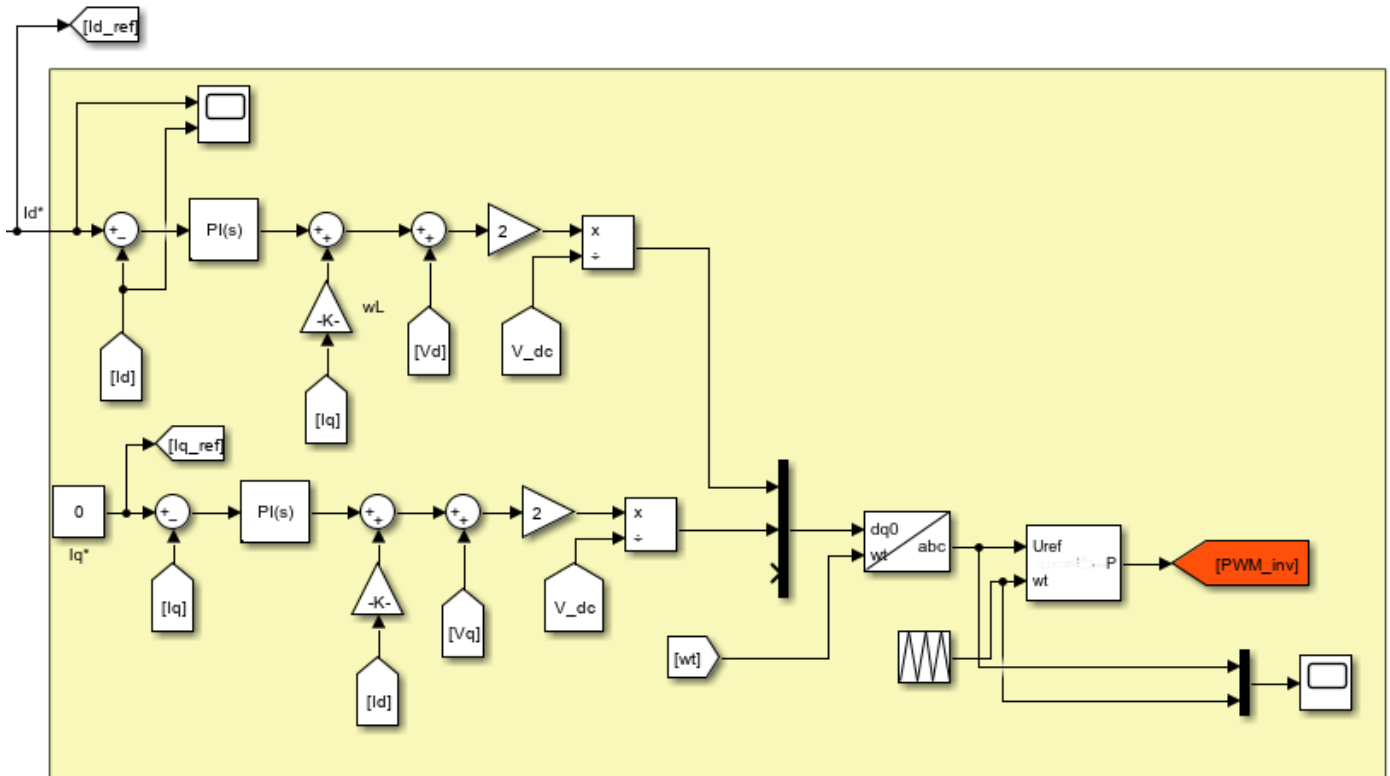


Figure 04: Current Loop Control (CLC) System

Methodology

The following signals feed into the CLC system:

- Measured currents (I_d , I_q) of the inverter.
- Reference currents (I_{d_ref} , I_{q_ref}) derived from the grid voltage.
- DC link voltage (V_{dc}) as input to the control system to maintain proper inverter voltage.

Control Algorithm

- Error Calculation: The system calculates the error between the measured currents and their respective reference currents:

$$e_d = I_{d_{ref}} - I_d, e_q = I_{q_{ref}} - I_q$$

- PI Controllers: The errors are processed by PI controllers for the d-axis and q-axis currents to generate the required voltage signals (Vd and Vq) that drive the inverter's output.
- PWM Generation: The voltage commands are converted into PWM signals to control the inverter's switching devices, ensuring that the inverter's current matches the reference values in both the d-axis (real power) and q-axis (reactive power).

d-q Transformation

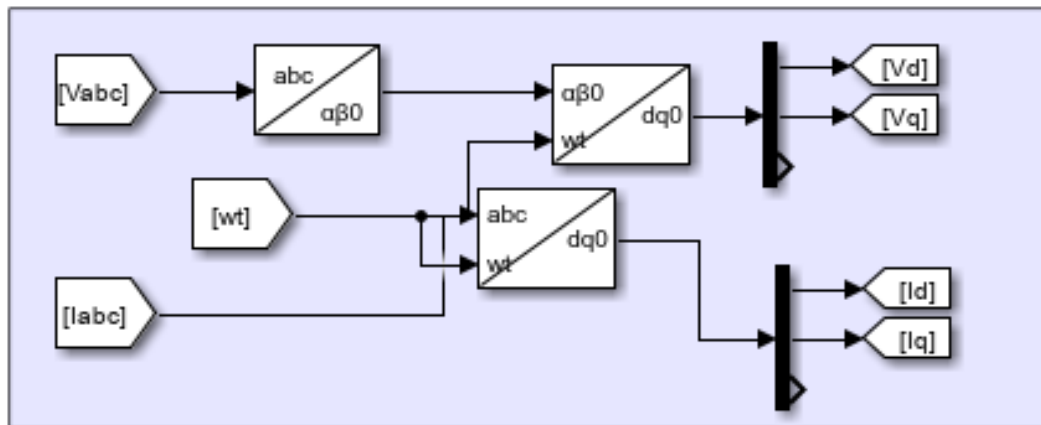


Figure 05: abc to d-q Transformation

- The abc to dq0 transformation ensures that the control loop operates in a rotating reference frame, which simplifies the separation of active and reactive power components.
- The system utilizes the dq0 transformation to convert the measured current from the three-phase system (abc) to a two-phase rotating system (dq), making the control system more effective.

5. MPPT OPTIMIZATION – ENHANCED P&O ALGORITHM

The MPPT Optimization focuses on enhancing the performance of the Perturb and Observe (P&O) algorithm for maximum power point tracking (MPPT) in solar PV systems. The P&O method adjusts the DC operating voltage of the solar PV array to extract the maximum power under varying irradiance conditions. However, the conventional P&O algorithm can exhibit slower response times and oscillations around the maximum power point (MPP). This report details the enhancements made to the P&O algorithm, optimizing its performance for faster convergence and reduced oscillations.

P&O Algorithm Explanation

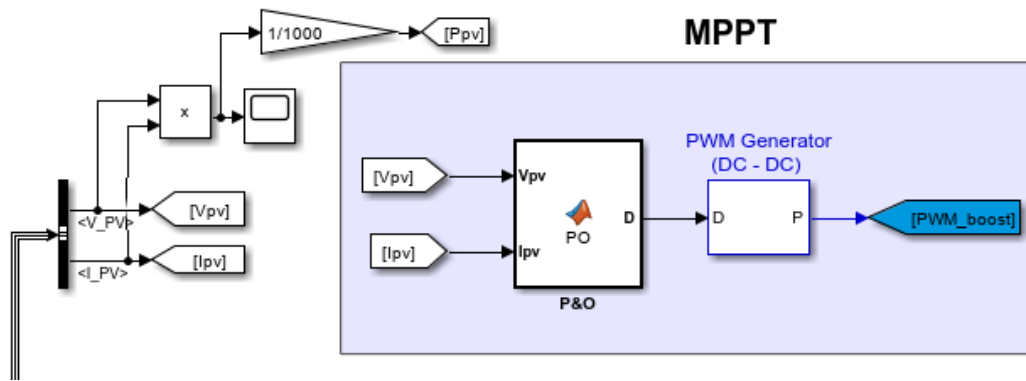


Figure 06: MPPT Control Block (P&O Algorithm)

The P&O MPPT algorithm works by continuously adjusting the operating voltage of the solar PV array and monitoring the resulting changes in power. The algorithm increases or decreases the operating voltage to find the maximum power point by observing the following:

1. **Voltage and Power Measurement:** The voltage (V_{pv}) and current (I_{pv}) of the PV array are measured, and the power (P_{pv}) is calculated as:

$$P_{pv} = V_{pv} \times I_{pv}$$

2. **Perturbation Logic:** The algorithm perturbs the voltage by a small increment (ΔD) and observes the resulting change in power:
 - If power increases, the system continues to perturb the voltage in the same direction.
 - If power decreases, the system reverses the direction of the perturbation.
3. **Control of the Operating Voltage:** The change in voltage is controlled within set limits (D_{min} and D_{max}) to ensure the voltage remains within an optimal range for the DC-DC converter.

Enhanced P&O Algorithm

The enhanced P&O algorithm improves the traditional P&O method by refining the perturbation steps based on power and voltage variations, ensuring faster tracking and fewer oscillations near the maximum power point (MPP).

Key Enhancements:

1. **Dynamic Perturbation Size:** The perturbation size (ΔD) is optimized to improve the tracking speed and minimize oscillations around the MPP.
2. **Boundary Checking:** The output reference voltage (D) is bounded between D_{min} and D_{max} to avoid instability caused by excessive perturbation.
3. **Initial Conditions:** The algorithm starts with initial values for D , V_{pre} , and P_{pre} to avoid sudden jumps in operation.

Algorithm Implementation

Here is the code for the Enhanced P&O MPPT Algorithm:

```

1  function D = P0(Vpv,Ipv)
2      Dmax=0.9;
3      Dmin=0.1;
4      Dinit=0.9;
5      deltaD=1e-3;
6      persistent Dpre Ppre Vpre;
7
8      %first run
9      if isempty(Dpre)
10         Dpre=Dinit;
11         Vpre=0;
12         Ppre=0;
13     end
14     Ppv=Vpv*Ipv;
15     dp=Ppv-Ppre;
16     dv=Vpv-Vpre;
17
18     if dp~=0
19         if(dp>0)
20             if(dv<0)
21                 D=Dpre+deltaD;
22             else
23                 D=Dpre-deltaD;
24             end
25         else
26             if(dv<0)
27                 D=Dpre-deltaD;
28             else
29                 D=Dpre+deltaD;
30             end
31         end
32     else
33         D=Dpre;
34     end
35
36     if D>=Dmax || D <= Dmin
37         D=Dpre;
38     end
39
40     Ppre=Ppv;
41     Vpre=Vpv;
42     Dpre=D;
43 end
```

Figure 07: Enhanced P&O MPPT Algorithm Code

6. FUTURE PLAN

- ✓ Performance Evaluation – Validate overall system efficiency under varying environmental conditions (irradiance, temperature, and shading).
- ✓ Power Quality Check – THD Analysis – Perform THD analysis to verify grid compliance with $\text{THD} < 5\%$.
- ✓ Final Report Preparation – Document system architecture, MPPT algorithm implementation, PLL & CLC integration, THD measurement, efficiency comparisons, and stability assessment.
- ✓ Presentation Slides – Prepare slides summarizing project overview, proposed architecture, MPPT algorithm comparisons, simulation results, THD analysis, and conclusions.

7. WEEK 13 SUMMARY

Table 02: Progress and Status

Topic	Description	Status
PLL Integration	PLL implemented for precise phase & frequency locking using dq-frame transformation.	Completed
CLC Implementation	dq-frame current loop control with PI regulators tested; inverter current tracking validated.	Completed
DC-Link Control	Outer voltage PI loop cascaded with CLC implemented; Vdc regulation at 600 V verified under varying irradiance.	Completed
MPPT Enhancement	Variable step P&O with global search phase implemented and tested under shading scenarios.	Completed
Efficiency Validation	Comprehensive efficiency testing across irradiance and temperature profiles planned for Week 14.	Planned
THD Analysis	FFT-based harmonic analysis targeting grid compliance ($\text{THD} < 5\%$) planned for Week 14.	Planned
Documentation	Final report and presentation slides for Week 15 submission – drafting to begin.	Planned